

Q (क)

$$\frac{d}{dx} \ln(\sec x + \tan x) = \frac{1}{\sec x + \tan x} \cdot (\sec x \tan x + \sec^2 x) = \sec x$$

2) $y = \ln(\sec x + \tan x)$

Diff. w.r. to x , (we get)

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} \\ &= \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x} = \sec x \end{aligned}$$

3) $y = \ln(\sqrt{x-a} + \sqrt{x-b})$

Diff. w.r. to x we get,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{x-a} + \sqrt{x-b}} \times \left[\frac{1}{2\sqrt{x-a}} + \frac{1}{2\sqrt{x-b}} \right] \\ &= \frac{1}{(\sqrt{x-a} + \sqrt{x-b})} \times \frac{1}{2} \frac{(\sqrt{x-b} + \sqrt{x-a})}{\sqrt{x-a} \cdot \sqrt{x-b}} = \frac{1}{2} \times \frac{1}{\sqrt{x-a} \sqrt{x-b}} \end{aligned}$$

4) $y = \ln(x + \sqrt{x^2 + a^2})$

Diff. w.r. to x we get,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{x + \sqrt{x^2 + a^2}} \left(1 + \frac{1}{2\sqrt{x^2 + a^2}} \cdot 2x \right) \\ &= \frac{1}{x + \sqrt{x^2 + a^2}} \times \left(\frac{\sqrt{x^2 + a^2} + x}{\sqrt{x^2 + a^2}} \right) \\ &= \frac{1}{\sqrt{x^2 + a^2}} \end{aligned}$$

81 $y = \ln \{x + \sqrt{x^2 + 2}\} + \sec^{-1} x^2 + \sqrt{5}$

Diff. w. r. to x we get,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{x + \sqrt{x^2 + 2}} \times \left(1 + \frac{2x}{2\sqrt{x^2 + 2}}\right) + \frac{2x}{x^2 \sqrt{x^2 - 1}} \\ &= \frac{1}{\sqrt{x^2 + 2}} + \frac{2}{x\sqrt{x^2 - 1}} \\ &= \frac{x\sqrt{x^2 - 1} + 2\sqrt{x^2 + 2}}{x\sqrt{x^2 - 1} \times \sqrt{x^2 + 2}} \end{aligned}$$

82 $y = \ln(1 + \sin x) - 2 \ln \sec\left(\frac{\pi}{4} - x\right)$

Diff. w. r. to x we get,

$$\begin{aligned} \frac{dy}{dx} &= \left[\frac{\cos x}{1 + \sin x} + \frac{2 \sec\left(\frac{\pi}{4} - x\right) \tan\left(\frac{\pi}{4} - x\right)}{\sec^2\left(\frac{\pi}{4} - x\right)} \right] \\ &= \frac{\cos x}{1 + \sin x} + 2 \tan\left(\frac{\pi}{4} - x\right) \end{aligned}$$

83 $y = \{1 + \tan(1 + \sqrt{x})\}^{1/3}$

Diff. w. r. to x we get,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{3} \left[1 + \tan(1 + \sqrt{x})\right]^{-2/3} \times \sec^2(1 + \sqrt{x}) \times \frac{1}{2\sqrt{x}} \\ &= \frac{\sec^2(1 + \sqrt{x})}{3 \left[1 + \tan(1 + \sqrt{x})\right]^{2/3} 2\sqrt{x}} \end{aligned}$$

Q1 $y = \ln \sin(x^2 + 5)$

Diff. w.r. to x we get,

$$\frac{dy}{dx} = \frac{\cos(x^2 + 5) \cdot 2x}{\sin(x^2 + 5)}$$

Q2 $y = \ln \sec(ax + b)^3$

Diff. w.r. to x we get,

$$\begin{aligned} \frac{dy}{dx} &= \frac{3 \sec(ax + b)^2 \sec(ax + b) \tan(ax + b)}{\sec(ax + b)^3} \\ &= 3 \tan(ax + b) \end{aligned}$$

Q3 $y = \ln \sec(ax + b)^3$

Diff. w.r. to x we get,

$$\frac{dy}{dx} = \frac{\sec(ax + b)^3 \tan(ax + b)}{\sec(ax + b)^3} \times 3(ax + b)^2 \cdot a$$

$$= 3a(ax + b)^2 \tan(ax + b)$$

Q4 $y = \sin^2(\ln x^2)$

$$\frac{dy}{dx} = 2 \sin(\ln x^2) \cos(\ln x^2) \cdot \frac{2x}{x^2}$$

$$= (\sin 2 \ln x^2) \frac{2}{x}$$

$$= (\sin 4 \ln x) \frac{2}{x}$$

$$17) y = \sec \left\{ \frac{1}{2} \ln(x^2 + a^2) \right\}$$

$$\frac{dy}{dx} = \sec \left\{ \frac{1}{2} \ln(x^2 + a^2) \right\} \tan \left\{ \frac{1}{2} \ln(x^2 + a^2) \right\} \cdot \frac{1}{2} \cdot \frac{2x}{(x^2 + a^2)}$$

$$18) y = \cos(\ln x) + \ln(\tan x)$$

$$\frac{dy}{dx} = -\frac{\sin(\ln x)}{x} + \frac{\sec^2 x}{\tan x}$$

$$= -\frac{\sin(\ln x)}{x} + \frac{\sin x}{\sin^2 x \cos x}$$

$$= -\frac{\sin(\ln x)}{x} + \frac{1}{2 \sin x \cos x}$$

$$= -\frac{\sin(\ln x)}{x} + 2 \operatorname{cosec} 2x$$

$$20) y = \ln [\sqrt{1+\ln x} - \sin x]$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1+\ln x} - \sin x} \times \left(\frac{1}{2\sqrt{1+\ln x}} - \cos x \right)$$

$$= \frac{1}{\sqrt{1+\ln x} - \sin x} \times \frac{1 - 2x \cos x \sqrt{1+\ln x}}{2x \sqrt{1+\ln x}}$$

$$21) y = \tan^{-1} \left(\frac{a+x}{1-ax} \right)$$

$$= \tan^{-1} a + \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$22) y = \ln \frac{\sqrt{x^2+1} - x}{\sqrt{x^2+1} + x}$$

$$y = \ln(\sqrt{x^2+1} - x) - \ln(\sqrt{x^2+1} + x)$$

$$\frac{dy}{dx} = \frac{\frac{2x}{2\sqrt{x^2+1}} - 1}{\sqrt{x^2+1} - x} - \frac{\frac{2x}{2\sqrt{x^2+1}} + 1}{\sqrt{x^2+1} + x}$$

$$= \frac{x - \sqrt{x^2+1}}{\sqrt{x^2+1}(\sqrt{x^2+1} - x)} - \frac{x + \sqrt{x^2+1}}{(\sqrt{x^2+1} + x)\sqrt{x^2+1}}$$

$$= -\frac{1}{\sqrt{x^2+1}} - \frac{1}{\sqrt{x^2+1}}$$

$$= -\frac{2}{\sqrt{x^2+1}}$$

$$22) y = \frac{1}{2} \ln(\sqrt{x^2+1} - 1) - \frac{1}{2} \ln(\sqrt{x^2+1} + 1)$$

$$\frac{dy}{dx} = \frac{\frac{2x}{2\sqrt{x^2+1}}}{2(\sqrt{x^2+1} - 1)} - \frac{\frac{2x}{2\sqrt{x^2+1}}}{2(\sqrt{x^2+1} + 1)}$$

$$= \frac{1}{2} \cdot \frac{x}{\sqrt{x^2+1}} \times \frac{\sqrt{x^2+1} + 1 - \sqrt{x^2+1} + 1}{x^2+1 - 1}$$

$$= \frac{1}{2} \cdot \frac{x}{\sqrt{x^2+1}} \times \frac{2}{x^2}$$

$$= \frac{1}{x\sqrt{x^2+1}}$$

$$= \frac{1}{x\sqrt{x^2+1}}$$

27) $y = x^n e^{\cos x}$

$$\begin{aligned} \frac{dy}{dx} &= nx^{n-1} e^{\cos x} + x^n e^{\cos x} (-\sin x) \\ &= nx^{n-1} e^{\cos x} - x^n \sin x e^{\cos x} \end{aligned}$$

28) $y = x e^{ax} \sin^m x$

$$\begin{aligned} \frac{dy}{dx} &= a e^{ax} \sin^m x + m x e^{ax} \sin^{m-1} x \cdot \cos x \\ &= e^{ax} \sin^m x (a + m \cot x) \end{aligned}$$

29) $y = \sin^m x \cos^n x$

$$\begin{aligned} \frac{dy}{dx} &= m \sin^{m-1} x \cos^n x \cdot \cos x - n \cos^{n-1} x \sin^m x \cdot \sin x \\ &= \sin^m x \cos^n x (m \cot x - n \tan x) \end{aligned}$$

30) $y = \frac{\sin^m x}{\cos^n x}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\cos^n x \cdot m \sin^{m-1} x \cdot \cos x + \sin^m x \cdot n \cos^{n-1} x \cdot \sin x}{\cos^{2n} x} \\ &= \frac{\cos^n x \sin^m x (m \cot x + n \tan x)}{\cos^{2n} x} \\ &= \frac{\sin^m x (m \cot x + n \tan x)}{\cos^n x} \end{aligned}$$

29/ $y = x \sec x \ln(xe^x)$

$$\begin{aligned} \frac{dy}{dx} &= x \sec x \cdot \frac{xe^x + e^x}{xe^x} + \sec x \ln(xe^x) + x \ln(xe^x) \sec x \tan x \\ &= \sec x (x+1) + \sec x \ln(xe^x) + x \ln(xe^x) \sec x \tan x \\ &= \sec x [(x+1) + \ln(xe^x) + x \ln(xe^x) \tan x] \end{aligned}$$

26/ $y = \cot^{-1} \frac{a}{x} + \ln \sqrt{\frac{x-a}{x+a}}$

$$y = \cot^{-1} \frac{a}{x} + \ln \sqrt{\frac{x-a}{x+a}}$$

$$\frac{dy}{dx} = -\frac{1}{1 + \frac{a^2}{x^2}} \cdot \frac{-a}{x^2} + \frac{1}{\sqrt{x-a}} \cdot \frac{1}{2\sqrt{x-a}} - \frac{1}{\sqrt{x+a}} \cdot \frac{1}{2\sqrt{x+a}}$$

$$= \frac{x^2}{x^2+a^2} \cdot \frac{a}{x^2} + \frac{1}{2(x-a)} - \frac{1}{2(x+a)}$$

$$= \frac{a}{x^2+a^2} + \frac{x+a-x+a}{2(x^2-a^2)} = \frac{2a}{x^2+a^2} + \frac{2a}{2(x^2-a^2)} = \frac{2a}{x^2-a^2}$$

$$= \frac{a}{x^2+a^2} + \frac{a}{x^2-a^2}$$

$$= \frac{ax^2 - a^3 + ax^2 + a^3}{x^4 - a^4}$$

$$= \frac{2ax^2}{x^4 - a^4}$$

$$20) \quad y = \ln\left(\frac{1+x}{1-x}\right)^{1/4} + \frac{1}{2} \tan^{-1} x^2$$

$$y = \frac{1}{4} \ln(1+x) - \frac{1}{4} \ln(1-x) + \frac{1}{2} \tan^{-1} x^2$$

$$\frac{dy}{dx} = \frac{1}{4} \cdot \frac{1}{1+x} + \frac{1}{4(1-x)} - \frac{1}{2} \times \frac{1}{1+x^2} \times 2x$$

$$= \frac{1}{4(1+x)} + \frac{1}{4(1-x)} - \frac{x}{1+x^2}$$

$$= \frac{1-x+1+x}{4(1+x^2)} - \frac{x}{1+x^2}$$

$$= \frac{1}{2(1+x^2)} - \frac{x}{1+x^2} = \frac{1-x}{1+x^2}$$

$$21) \quad y = \sin^{-1} x^2 - x e^x$$

$$\frac{dy}{dx} = \frac{2x}{\sqrt{1-x^2}} - e^x - x e^x$$

$$= \frac{2x}{\sqrt{1-x^2}} - e^x (1+x)$$

$$22) \quad y = (1+x) \tan^{-1} \sqrt{x} - \sqrt{x}$$

$$\frac{dy}{dx} = \frac{1+x}{1+x} \times \frac{1}{2\sqrt{x}} + \tan^{-1} \sqrt{x} - \frac{1}{2\sqrt{x}}$$

$$= \tan^{-1} \sqrt{x}$$

$$20) \quad y = \ln \frac{x}{(1+x)^{1/2} (1+x^2)^{1/4}} - \left(\frac{1}{2} \tan^{-1} x \right) \quad \text{Ans}$$

$$= \ln x - \frac{1}{2} \ln(1+x) - \frac{1}{4} \ln(1+x^2) - \frac{1}{2} \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{1}{x} - \frac{1}{2(1+x)} - \frac{2x}{4(1+x^2)} - \frac{1}{2(1+x^2)}$$

$$= \frac{(1+x)(1+x^2) - 2x(1+x^2) - (1+x)(1+x^2)}{2x(1+x)(1+x^2)} \quad \text{Ans}$$

$$= \frac{1}{x} - \frac{1}{2(1+x)} - \frac{x}{2(1+x^2)} - \frac{1}{2(1+x^2)}$$

$$= \frac{2(1+x)(1+x^2) - x(1+x^2) - x^2(1+x) - x(1+x)}{2x(1+x)(1+x^2)} \quad \text{Ans}$$

$$= \frac{2 + 2x^2 + 2x + 2x^3 - x - x^3 - x^2 - x^3 - x - x^3}{2x(1+x)(1+x^2)}$$

$$= \frac{1}{x(1+x)(1+x^2)} \quad \text{Ans}$$

$$22) \quad y = (x^2+1) \sin^{-1} x + e^{\sqrt{1+x^2}}$$

$$\frac{dy}{dx} = 2x \sin^{-1} x + \frac{x^2+1}{\sqrt{1-x^2}} + e^{\sqrt{1+x^2}} \cdot \frac{1}{2\sqrt{1+x^2}} \cdot 2x$$

$$= 2x \sin^{-1} x + \frac{x^2+1}{\sqrt{1-x^2}} + x e^{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}}$$

$$\underline{28]} \quad y = (a^{2/3} - x^{2/3})^{3/2}$$

$$\frac{dy}{dx} = \frac{3}{2} (a^{2/3} - x^{2/3})^{1/2} \times -\frac{2}{3} x^{-1/3}$$

$$= -\frac{1}{x^{1/3}} \sqrt{a^{2/3} - x^{2/3}}$$

$$\underline{29]} \quad y = \tan^{-1} \frac{a^{1/3} + x^{1/3}}{1 - a^{1/3} x^{1/3}}$$

$$\frac{dy}{dx} = \frac{1}{1 + \frac{(a^{1/3} + x^{1/3})^2}{(1 - a^{1/3} x^{1/3})^2}} \times \frac{1}{3} x^{-2/3} (1 - a^{1/3} x^{1/3}) - \frac{1}{3} x^{-2/3} (a^{1/3} + x^{1/3})$$

$$y = \tan^{-1} a^{1/3} + \tan^{-1} x^{1/3}$$

$$\frac{dy}{dx} = \frac{1}{1 + x^{2/3}} \times \frac{1}{3} x^{-2/3}$$

$$= \frac{1}{3x^{3/2}} \times \frac{1}{1 + x^{2/3}}$$

$$\underline{29]} \quad y = \tan^{-1} \frac{x}{\sqrt{1-x^2}} \quad \text{let,}$$

$$x = \sin \theta$$

$$y = \tan^{-1} \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$$

$$\theta = \sin^{-1} x$$

$$y = \tan^{-1} \frac{\sin \theta}{\cos \theta}$$

$$y = \tan^{-1} \tan \theta = \theta = \sin^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$26) y = \cos^{-1} \frac{1-x^2}{1+x^2}$$

$$= 2 \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{2}{1+x^2}$$

$$27) y = \sin^{-1} \frac{x}{\sqrt{1+x^2}}$$

$$= \sin^{-1} \frac{\tan \theta}{\sqrt{1+\tan^2 \theta}}$$

$$= \sin^{-1} \frac{\tan \theta}{\sec \theta}$$

$$= \sin^{-1} \frac{\sin \theta}{\cos \theta} \cdot \cos \theta$$

$$\frac{1}{x+1} \times \sin^{-1} \sin \theta = \frac{\theta}{x+1}$$

$$y = \theta = \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$28) y = \tan^{-1} \frac{\sqrt{1+x^2} - 1}{x}$$

$$= \tan^{-1} \frac{\sqrt{1+\tan^2 \theta} - 1}{\tan \theta}$$

$$= \tan^{-1} \frac{\sec \theta - 1}{\tan \theta}$$

$$= \tan^{-1} \frac{1 - \cos \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta}$$

$$= \tan^{-1} \frac{1 - \cos \theta}{\sin \theta}$$

$$\frac{\sin \theta \cos \theta}{1 - \cos \theta} = \frac{1}{2}$$

$$x^2 - \cos \theta = \frac{1}{2}$$

$$\frac{1}{x+1} \times \frac{1}{x} = \frac{1}{x(x+1)}$$

Let,

$$\frac{x}{x+1} = \tan \theta = \frac{1}{x}$$

$$x^2 - 1 = \tan^{-1} x$$

$$\frac{x}{x+1} = \tan \theta$$

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$$\frac{x}{x+1} = \tan \theta$$

$$\frac{x}{x+1} = \tan \theta$$

$$\frac{x}{x+1} = \tan \theta$$

$$x = \tan \theta$$

$$\theta = \tan^{-1} x$$

$$x = \tan \theta$$

$$\theta = \tan^{-1} x$$

$$\frac{x}{x+1} = \tan \theta$$

$$y = \tan^{-1} \frac{2\sin^2 \theta/2}{1 - 2\sin \theta/2 \cos \theta/2}$$

$$= \theta/2 = \frac{1}{2} \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \times \frac{1}{1+x^2}$$

$$\text{Q21 } y = \tan^{-1} \frac{4\sqrt{x}}{1-4x}$$

$$= \tan^{-1} \frac{2 \cdot 2\sqrt{x}}{1-(2\sqrt{x})^2}$$

$$= 2 \tan^{-1} (2\sqrt{x})$$

$$\therefore \frac{dy}{dx} = \frac{2}{1+(2\sqrt{x})^2} \times \frac{2}{2\sqrt{x}} = \frac{2}{1+4x} \times \frac{1}{\sqrt{x}}$$

$$\text{Q21 } y = \sin^{-1} \frac{1-x^2}{1+x^2}$$

$$= \sin^{-1} \frac{1-\tan^2 \theta}{1+\tan^2 \theta}$$

$$= \sin^{-1} \cos 2\theta$$

$$= \sin^{-1} \sin(\frac{\pi}{2} - 2\theta)$$

$$= \frac{\pi}{2} - 2\theta$$

$$\therefore \frac{dy}{dx} = -\frac{2}{1+x^2}$$

$$\text{Q77: } y = \cos^{-1} \{ 2x\sqrt{1-x^2} \}$$

Let, $x = \sin \theta$

$\frac{1}{s} \cdot \theta = \sin^{-1} x$

$$y = \cos^{-1} \{ 2 \sin \theta \sqrt{1 - \sin^2 \theta} \}$$

$$= \cos^{-1} \{ 2 \sin \theta \cos \theta \}$$

$$= \cos^{-1} \sin 2\theta$$

$$= \cos^{-1} \cos \left(\frac{\pi}{2} - 2\theta \right)$$

$$= \frac{\pi}{2} - 2\theta$$

$$= \frac{\pi}{2} - 2 \sin^{-1} x$$

$$\frac{dy}{dx} = \frac{-2}{\sqrt{1-x^2}}$$

$$\text{Q81: } y = \cot^{-1} \{ \sqrt{1+x^2} - x \}$$

$$= \cot^{-1} \{ \sqrt{1+\cot^2 \theta} - \cot \theta \}$$

$$= \cot^{-1} (\operatorname{cosec} \theta - \cot \theta)$$

$$= \cot^{-1} \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)$$

$$= \cot^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right)$$

$$= \cot^{-1} \frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \cot^{-1} \tan \frac{\theta}{2}$$

$$= \cot^{-1} \cot \left(\frac{\pi}{2} - \frac{\theta}{2} \right)$$

$$\text{or } y = -\frac{\theta}{2}$$

$$\text{or } y = -\frac{1}{2} \cot^{-1} x$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2} \left(-\frac{1}{1+x^2} \right)$$

$$= \frac{1}{2} \times \frac{1}{1+x^2}$$

$$\text{Q6)} \quad y = \tan^{-1} \frac{\sqrt{1-\cos x}}{\sqrt{1+\cos x}}$$

$$= \tan^{-1} \frac{\sqrt{2\sin^2 \frac{x}{2}}}{\sqrt{2\cos^2 \frac{x}{2}}}$$

$$= \tan^{-1} \tan \frac{x}{2}$$

$$= \frac{x}{2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2}$$

$$\text{Q7)} \quad y = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$$

$$\text{let, } x = \cos \theta$$

$$\theta = \cos^{-1} x$$

$$y = \tan^{-1} \sqrt{\frac{1-\cos \theta}{1+\cos \theta}}$$

$$= \tan^{-1} \sqrt{\frac{2\sin^2 \frac{\theta}{2}}{2\cos^2 \frac{\theta}{2}}}$$

$$\theta = \frac{\pi}{2}$$

$$= \tan^{-1} \tan \frac{\theta}{2}$$

$$\therefore y = \frac{\theta}{2} = \frac{1}{2} \cos^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \frac{1}{\sqrt{1-x^2}}$$

$$\text{Q9)} \quad y = \tan^{-1} \frac{\cos x - \sin x}{\cos x + \sin x}$$

$$= \tan^{-1} \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}}$$

$$= \tan^{-1} \frac{1 - \tan x}{1 + \tan x}$$

$$= \tan^{-1} \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \cdot \tan x}$$

$$= \tan^{-1} \tan \left(\frac{\pi}{4} - x \right)$$

$$\therefore y = \frac{\pi}{4} - x$$

$$\frac{dy}{dx} = -1$$

$$\text{Q6)} \quad y = \tan^{-1} \frac{1 + \tan x}{1 - \tan x}$$

$$= \tan^{-1} \frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \cdot \tan x}$$

$$= \tan^{-1} \tan \left(\frac{\pi}{4} + x \right)$$

$$= \frac{\pi}{4} + x$$

$$\therefore \frac{dy}{dx} = 1$$

$$x = \frac{\pi}{4} - \frac{\pi}{4} = \frac{\pi}{4} - \frac{\pi}{4} = 0$$

$$\frac{1}{x-1} \times \frac{1}{5} = \frac{66}{x-1}$$

$$\text{Q21 } y = \tan^{-1} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \quad \text{let, } x = \cos \theta$$

$$y = \tan^{-1} \frac{\sqrt{1+\cos \theta} - \sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta} + \sqrt{1-\cos \theta}} \quad \theta = \cos^{-1} x$$

$$= \tan^{-1} \frac{\sqrt{2} \cos \frac{\theta}{2} - \sqrt{2} \sin \frac{\theta}{2}}{\sqrt{2} \cos \frac{\theta}{2} + \sqrt{2} \sin \frac{\theta}{2}}$$

$$= \tan^{-1} \frac{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}$$

$$= \tan^{-1} \frac{1 - \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}}{1 + \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}}$$

$$= \tan^{-1} \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}$$

$$= \tan^{-1} \frac{\tan \frac{\pi}{4} - \tan \frac{\theta}{2}}{1 + \tan \frac{\pi}{4} \cdot \tan \frac{\theta}{2}}$$

$$= \tan^{-1} \tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right)$$

$$= \frac{\pi}{4} - \frac{\theta}{2} = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \times \frac{1}{\sqrt{1-x^2}}$$

801 $y = \tan^{-1} \frac{\cos x}{1 + \sin x}$

$\theta + 0 = x$
 $x = \theta \Rightarrow \tan^{-1}$

$$\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$1 + \frac{2 \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$= \tan^{-1} \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \times \frac{1 + \tan^2 \frac{x}{2}}{1 + 2 \tan^2 \frac{x}{2} + \tan^2 \frac{x}{2}}$$

$$= \tan^{-1} \frac{1 - \tan^2 \frac{x}{2}}{(1 + \tan^2 \frac{x}{2})^2}$$

$$= \tan^{-1} \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$= \tan^{-1} \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$= \frac{\pi}{4} - \frac{x}{2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2}$$

$$82) y = \sec^{-1} \frac{x^2+1}{x^2-1}$$

$$= \sec^{-1} \frac{\cot^2 \theta + 1}{\cot^2 \theta - 1}$$

$$= \sec^{-1} \frac{1 + \tan^2 \theta}{\tan^2 \theta} \times \frac{\tan \theta}{1 - \tan^2 \theta}$$

$$= \sec^{-1} \frac{1 + \tan^2 \theta}{1 - \tan^2 \theta}$$

$$= \cos^{-1} \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$= \cos^{-1} \cos 2\theta$$

$$= 2\cot^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{-2}{1+x^2}$$

$$\text{let } \cot \theta = x \quad \text{102}$$

$$x = \cot \theta$$

$$\theta = \cot^{-1} x$$

$$\frac{x}{s} = \frac{p}{p}$$

$$\frac{1}{s} = \frac{b}{x b} \therefore$$

$$821 \quad y = \tan^{-1} \frac{a+bx}{b-ax}$$

$$= \tan^{-1} \frac{\frac{a}{b} + \frac{b}{b}x}{\frac{b}{b} - \frac{a}{b}x}$$

$$= \tan^{-1} \frac{\frac{a}{b} + x}{1 - \frac{a}{b}x}$$

$$= \tan^{-1} \frac{a}{b} + \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$801 \quad y = \tan^{-1} \left\{ \sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} \right\}$$

$$\frac{dy}{dx} = \frac{1}{1 + \frac{a-b}{a+b} \tan^2 \frac{x}{2}} \cdot \sec^2 \frac{x}{2} \times \frac{1}{2}$$

$$= \frac{a+b}{(a+b) + (a-b) \tan^2 \frac{x}{2}} \cdot \frac{1}{2} \cdot \frac{\sqrt{a-b}}{\sqrt{a+b}} \cdot \frac{1}{\cos^2 \frac{x}{2}}$$

$$= \frac{\cos^2 \frac{x}{2} (a+b)}{\cos^2 \frac{x}{2} (a+b) + (a-b) \sin^2 \frac{x}{2}} \times \frac{1}{2} \cdot \frac{\sqrt{a-b}}{\sqrt{a+b}} \cdot \frac{1}{\cos^2 \frac{x}{2}}$$

$$= \frac{\sqrt{a+b} \sqrt{a-b}}{2 \{ a (\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}) + b (\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}) \}}$$

$$= \frac{\sqrt{a^2 - b^2}}{2(a + b \cos x)}$$

$$88) y = \cos^{-1} \frac{a+b\cos x}{b+a\cos x}$$

Putting, $z = \frac{a+b\cos x}{b+a\cos x}$ $\therefore y = \cos^{-1} z$

$$\frac{dy}{dz} = - \frac{1}{\sqrt{1-z^2}}$$

$$z = \frac{a+b\cos x}{b+a\cos x}$$

$$\begin{aligned} \frac{dz}{dx} &= \frac{(b+a\cos x)(-\sin x) + (a+b\cos x)a\sin x}{(b+a\cos x)^2} \\ &= \frac{-b^2\sin x - ab\sin x\cos x + a^2\sin x + ab\sin x\cos x}{(b+a\cos x)^2} \\ &= \frac{(a^2-b^2)\sin x}{(b+a\cos x)^2} \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= - \frac{1}{\sqrt{1-\left(\frac{a+b\cos x}{b+a\cos x}\right)^2}} \\ &= - \frac{(b+a\cos x)}{\sqrt{(b+a\cos x)^2 - (a+b\cos x)^2}} \end{aligned}$$

$$= - \frac{(b + a \cos x)}{\sqrt{b^2 + 2ab \cos x + a^2 \cos^2 x - a^2 - 2ab \cos x - b^2 \cos^2 x}}$$

$$= - \frac{(b + a \cos x)}{\sqrt{(b^2 - a^2) - (b^2 - a^2) \cos^2 x}}$$

$$= - \frac{(b + a \cos x)}{\sin x \sqrt{b^2 - a^2}}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

$$= \frac{-(b + a \cos x)}{\sin x \sqrt{b^2 - a^2}} \times \frac{-(b^2 - a^2) \sin x}{(b + a \cos x)^2}$$

$$= \frac{\sqrt{b^2 - a^2}}{b + a \cos x}$$

$$\frac{(b^2 - a^2) \sin x}{(b + a \cos x)^2} \times \frac{p + a \cos x}{\sin x \sqrt{b^2 - a^2}} = \frac{b}{x b}$$

$$83) \quad y = \sin^{-1} \frac{a+b\cos x}{b+a\cos x}$$

let,

$$z = \frac{a+b\cos x}{b+a\cos x}$$

$$\frac{dz}{dx} = \frac{(a^2 - b^2) \sin x}{(b+a\cos x)^2}$$

$$y = \sin^{-1} z$$

$$\frac{dy}{dz} = \frac{1}{\sqrt{1-z^2}}$$

$$= \frac{1}{\sqrt{1 - \frac{(a+b\cos x)^2}{(b+a\cos x)^2}}}$$

$$= \frac{b+a\cos x}{\sqrt{(b+a\cos x)^2 - (a+b\cos x)^2}}$$

$$= \frac{b+a\cos x}{\sin x \sqrt{b^2 - a^2}}$$

$$\therefore \frac{dy}{dx} = \frac{b+a\cos x}{\sin x \sqrt{b^2 - a^2}} \times \frac{(a^2 - b^2) \sin x}{(b+a\cos x)^2}$$

$$= \frac{-\sqrt{b^2 - a^2}}{b+a\cos x}$$

$$\underline{84)} \quad y = \sin^{-1} \left(\frac{2x^{-1}}{x+x^{-1}} \right)$$

$$= \sin^{-1} \frac{\frac{2}{x}}{x + \frac{1}{x}}$$

$$= \sin^{-1} \frac{2}{x} \times \frac{x}{x^2+1}$$

$$= \sin^{-1} \frac{2}{x^2+1}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - \frac{4}{(x^2+1)^2}}} \times 2x \times -\frac{2x}{(x^2+1)^2}$$

$$= \frac{x^2+1}{\sqrt{x^4+2x^2+1-4}} \times -\frac{4x}{(x^2+1)^2}$$

$$= \frac{\cancel{4(x^2+1)}}{\cancel{x} \sqrt{\cancel{x^4} + 2x^2 - 3}}$$

$$= \frac{-4x}{(x^2+1) \sqrt{x^4+2x^2-3}}$$